

Yichao Cai

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Research Interests

Representation learning theory; foundation-model objectives; multimodal learning; identifiability and representation geometry; reliable and interpretable machine learning.

Research Statement

My research studies how modern learning objectives and supervision signals shape learned representations. I am particularly interested in when objectives such as contrastive learning, masked prediction, and next-token prediction identify latent structure, and when they instead discard, conflate, or leave such structure underdetermined. Understanding these questions helps characterize the theoretical limits of foundation-model objectives, and distinguish which capabilities may emerge through scaling from which limitations require new objectives, supervision forms, or data interventions.

Methodologically, I use tools from identifiability theory, latent-variable modeling, population-objective analysis, and representation geometry. My broader goal is to develop a theory of representation learning that explains the capabilities and structural limits of multimodal foundation models, vision-language models, and predictive world models.

Education

Ph.D. in Computer Science

Sep 2023 – Mar 2027 expected

Australian Institute for Machine Learning, Adelaide University

Supervisor: Prof. Javen Qinfeng Shi

Expected thesis submission: Jan–Mar 2027

M.Sc. in Instrument Science and Technology

Sep 2016 – Jun 2019

Wuhan University of Technology

Supervisor: Prof. Xiao Zhou

B.Eng. in Measurement and Control Technology and Instrumentation

Sep 2012 – Jun 2016

Wuhan University of Technology

Publications

First-author and Co-first-author Publications

- [1] Cai, Y.; Zhang, Z.; Liu, Y.; and Shi, J. Q. "The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence." *International Conference on Machine Learning (ICML)*, 2026.
- [2] Cai, Y.*; Liu, Y.*; Gao, E.; Jiang, T.; Zhang, Z.; van den Hengel, A.; and Shi, J. Q. "On the Value of Cross-Modal Misalignment in Multimodal Representation Learning." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. **Spotlight**.
*Equal contribution.
- [3] Cai, Y.; Liu, Y.; Zhang, Z.; and Shi, J. Q. "CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts." *European Conference on Computer Vision (ECCV)*, 2024.

Collaborative Publications

- [4] Liu, Y.; Gong, D.; Cai, Y.; Gao, E.; Zhang, Z.; Huang, B.; Gong, M.; van den Hengel, A.; and Shi, J. Q. "I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?" *International Conference on Learning Representations (ICLR)*, 2026.
- [5] Jiang, W.; Liu, Y.; Cai, Y.; Gao, E.; Dong, J.; Abbasnejad, E.; Yao, L.; and Shi, J. Q. "What Makes a Representation Good for Single-Cell Perturbation Prediction?" *International Conference on Machine Learning (ICML)*, 2026.
- [6] Chen, J.; Yin, Z.; Cai, Y.; Liu, Y.; Zhang, Z.; Gong, D.; and Shi, J. Q. "Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement." *International Conference on Machine Learning (ICML)*, 2026.

Selected Research Projects

Semantic Identifiability under Cross-modal Misalignment

2024 – 2025

NeurIPS 2025 Spotlight, first/co-first author

- Developed a latent-variable model of multimodal contrastive learning under incomplete and biased language supervision, formalizing semantic selection bias and text perturbation bias in image-text supervision.
- Proved identifiability conditions characterizing which semantic factors are preserved, ignored, or learned as invariances, with implications for multimodal pretraining and robust representation learning.
- Validated the analysis with synthetic data, MPI3D-Complex, and OpenCLIP probing.

Geometric Mechanisms of Contrastive Representation Learning

2025 – 2026

ICML 2026, first author

- Developed a population-level geometric analysis of the InfoNCE objective, characterizing alignment potentials, entropy-driven dispersion, and cross-modal divergence.
- Explained how alignment, uniformity, and modality gap emerge from the contrastive objective, with implications for multimodal representation diagnostics and objective design.

Language-supervised Content–Style Disentanglement

2023 – 2024

ECCV 2024, first author

- Proposed CLAP, a contrastive learning framework that uses augmented language prompts to separate content and style factors in representation space, improving the cross-domain generalization and robustness of CLIP-like vision-language representations.
- Led method design, implementation, experiments, visualization, paper writing, and code release.

Ongoing Research

- **Masked prediction under metastable data laws:** studying when masked-prediction objectives, such as those used in MAE, BERT, JEPA, and masked diffusion models, identify the joint data law, and when they exhibit mode-blindness under slowly mixing or metastable structure.
- **Predictive-state identifiability in next-token predictors:** investigating what predictive-state information finite-capacity next-token predictors must retain, and which aspects of latent structure remain underdetermined by the training objective.
- **Objective-induced equivalence classes:** developing a framework for characterizing which representation structures are recoverable from data, supervision, and learning objectives.

Teaching

All at Adelaide University, formerly the University of Adelaide

Guest Lecturer and Head Tutor, Statistical Machine Learning	Semester 2, 2025
Teaching Assistant, Neural Networks and Deep Learning	Semester 1, 2026
Teaching Assistant, Using Machine Learning Tools	Trimester 2, 2025
Teaching Assistant, Concepts in AI and ML	Semester 1, 2025

Academic Service

Conference Reviewing: NeurIPS 2026; ICML 2026 (*Silver Reviewer Award*); ICLR 2026.

Journal Reviewing: Transactions on Machine Learning Research (TMLR).

Honors and Awards

The D. R. Stranks Travelling Fellowship	Jun 2026
NeurIPS 2025 Scholar Award, Neural Information Processing Foundation	Oct 2025
University of Adelaide Research Scholarship	Sep 2023
Outstanding Graduate Student, Wuhan University of Technology	Jun 2019

Additional Research Experience

Visiting Student Researcher

May 2018 – Sep 2018

California PATH, UC Berkeley

- Conducted research on visual perception algorithms for autonomous driving.